

◆ A.10 — H_∞ Robustness (Final — Rigorous Version)

A.10.1 System with Disturbance

$$\dot{\Psi}(t) = F(\Psi(t)) + B(u(t)) + d(t)$$

where:

- $d(t)$ is an exogenous disturbance, $d \in \mathcal{L}_2$
- $\|d(t)\|$ is bounded

A.10.2 Error Dynamics

Let Ψ^* be an equilibrium and define:
 $e(t) = \Psi(t) - \Psi^*$

Linearizing locally around Ψ^* :
 $\dot{e}(t) = A e(t) + B u(t) + d(t)$

A.10.3 State Feedback Control

$$u(t) = -K e(t)$$

Thus:

$$\dot{e}(t) = (A - BK)e(t) + d(t)$$

Denote $A_c := A - BK$.

A.10.4 Lyapunov Function

$$V(e) = e^T P e, \quad P \succ 0$$

A.10.5 Time Derivative

$$\begin{aligned} \dot{V} &= \\ e^T (A_c^T P + P A_c) e + 2 e^T P d \end{aligned}$$

A.10.6 Disturbance Bounding

Using Young's inequality, for any $\epsilon > 0$:

$$2 e^T P d$$

$\leq$

$$\epsilon e^T P e$$

$$+ \epsilon^{-1} d^T P d$$

A.10.7 Dissipation (Riccati/LMI Condition)

Assume there exist $P \succ 0$ and $Q \succ 0$ such that:

$$A_c^T P + P A_c \preceq -Q$$

Then:

$$\dot{V}$$

$$\leq -e^T Q e + \epsilon e^T P e + \epsilon^{-1} d^T P d$$

Choose ϵ small enough so that:

$$Q - \epsilon P \succeq \alpha I, \quad \alpha > 0$$

Hence:

$$\dot{V}$$

$$\leq -\alpha \|e\|^2 + \epsilon^{-1} \lambda_{\max}(P) \|d\|^2$$

A.10.8 H_∞ Gain (L_2 Stability)

Integrating over $[0,T]$:

$$\int_0^T \|e(t)\|^2 dt$$

\le

$$\frac{\lambda_{\max}(P)}{\alpha \, \varepsilon}$$

$$\int_0^T \|d(t)\|^2 dt$$

$$+ \frac{V(0)}{\alpha}$$

Define:

$$\gamma^2 := \frac{\lambda_{\max}(P)}{\alpha}$$

\varepsilonpsilon}

Then:

$$\|e\|_{\mathcal{L}_2}$$

\le

$$\gamma_{d|_{\mathcal{L}_2}}$$

$$+ \sqrt{\frac{V(0)}{\alpha}}$$

For zero initial condition:

\boxed{

$$|e|_{\mathcal{L}_2} \leq \gamma |d|_{\mathcal{L}}$$

$$_2\}$$

}

FINAL RESULT — A.10

$\boxed{}$

$\text{The closed-loop system is } \mathcal{H}_{\infty}\text{-stable with finite gain } \gamma.$

SEKHEM-TSDD Theory — Mohamed Abd Elnaby
DOI: 10.5281/zenodo.19521219